

Last updated:
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VINCENZO (VINNY) R. D'ONOFRIO
NRAO STUDENT OBSERVING SUPPORT PROGRAM FELLOW
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Research focus: galaxy formation and evolution across cosmic time with multiwavelength observations

EDUCATION

Texas A&M University

Ph.D. Astronomy

Advisor: Dr. Justin Spilker

Thesis: Tracing Galaxy Evolution Across Cosmic Time with Post-Starburst Galaxies

College Station, TX

Expected May 2027

M.S. Astronomy

May 2025

GPA: 3.92/4.00

Project: Quenching via Tidal Removal of Cold Gas in Intermediate-z Massive Post-Starburst Galaxies

Michigan State University

B.S. Astrophysics

GPA: 3.92/4.00 | Majors: Astrophysics, Physics

East Lansing, MI

May 2022

RESEARCH POSITIONS/EXPERIENCE

Graduate Student (Advisor: Dr. Justin Spilker)

Texas A&M University

August 2022 – Present

College Station, TX

Student Research Assistant (Advisor: Dr. Laura Chomiuk)

Michigan State University Observatory Research Program

August 2021 – May 2022

East Lansing, MI

NSF REU Intern (Advisor: Dr. Juergen Ott)

National Radio Astronomy Observatory

May 2021 – October 2021

Socorro, NM (Remote)

Student Research Assistant (Advisor: Dr. Kaitlin Cook)

National Superconducting Cyclotron Laboratory

June 2020 – May 2022

East Lansing, MI

Research Assistant (Advisor: Dr. Daniel Rathbun)

Bionics and Vision Lab, Henry Ford Health System

April 2019 – August 2019

Detroit, MI

HONORS AND AWARDS

TAMU Dept. of Physics & Astronomy Travel Award, **\$1k**

January 2026

NRAO Student Observing Support Program Fellowship, **\$40k**

November 2024

NSF Graduate Research Fellowship Program, **Honorable Mention**

April 2024

Texas Space Grant Consortium Graduate Fellowship, **\$5k**

December 2023

NSF Research Experiences for Undergraduates, **~\$7k**

June 2021

MSU College of Natural Science Undergraduate Research Support Scholarship, **\$1k** January 2021

AWARDED PROPOSALS

Principal Investigator

5. **ALMA Cycle 13** - 2026.1.00767.S (21.9 hours): The Fate of Molecular Gas After the Burst: Tracing the Dense Gas Content in $z \sim 0.7$ Post-Starburst Galaxies (PI: **V. D'Onofrio**)
4. **VLA Semester 26A** - VLA/26A-427 (21.0 hours): What Powers the Radio? Morphological Constraints in High- z Post-Starbursts (PI: **V. D'Onofrio**)
3. **ALMA Cycle 12** - 2025.1.01006.S (12.7 hours): What's Left Behind: A Census of the Cold ISM in the First Massive Quiescent Galaxies (PI: **V. D'Onofrio**)
2. **ALMA Cycle 11** - 2024.1.01252.S (9.8 hours): Diffuse or Dense: Probing the Physical State of Massive Gas Reservoirs in $z \sim 0.7$ Quenched Galaxies (PI: **V. D'Onofrio**)
1. **VLA Semester 24B** - VLA/24B-451 (21.0 hours): Measuring Jet Ages to Test Radio AGN Feedback with Massive $z \sim 0.7$ Post-Starbursts (PI: **V. D'Onofrio**)

Co-Investigator

10. **ALMA Cycle 13** - 2026.1.01631.S (13.2 hours): Cracking the depletion time problem in CO-luminous post-starburst galaxies: ^{13}CO as a test of the CO-to- H_2 conversion (PI: D. Setton)
9. **ALMA Cycle 13** - 2026.1.01360.S (16.7 hours): Quenching, Fast and Slow: Molecular Gas in Green Valley Galaxies at $z \sim 0.6$ (PI: K. Suess)
8. **ALMA Cycle 13** - 2026.1.00570.S (30.3 hours): $[\text{Cl}]+\text{CO}$ Diagnostics of the Circumgalactic Medium of GS-9209, a Massive Quiescent Galaxy at $z=4.658$ (PI: F. Sun)
7. **Roman Cycle 1** - Proposal ID: 10244, Program ID: 19068 (Program Category: Small): Connecting Morphological Transformation to Rapid Quenching: Resolving the Structure of $z \sim 1$ Post-Starburst Galaxies with Roman (PI: J. Spilker)
6. **VLA Semester 26A** - VLA/26A-430 (45.0 hours): Are High-Redshift Quiescent Galaxies Gas-Rich but Dust-Poor? (PI: J. Spilker)
5. **ALMA Cycle 12** - 2025.1.01067.S (45.6 hours): Dust Rising from the ASHES: A Comprehensive ALMA Survey of Remnant Dust in the Earliest Quiescent Galaxies (PI: Z. Ji)

4. **ALMA Cycle 11 w/ JWST Cycle 3** - 2024.1.01064.S/JWST-GO-06719 (15.5/4.7 hours): Mapping Cold Gas and Star Formation in Gas Rich Post-Starburst Galaxies Near Cosmic Noon (PI: D. Setton)
3. **ALMA Cycle 11** - 2024.1.00216.S (13.1 hours): Timing the Onset of Unexpected Dust Destruction using High-Redshift Post-Starburst Galaxies (PI: J. Spilker)
2. **ALMA Cycle 10** - 2023.1.01012.S (12.1 hours): Does Molecular Gas Survive Quenching Near Cosmic Noon? (PI: D. Setton)
1. **ALMA Cycle 10** - 2023.1.00948.S (13.1 hours): Timing the Onset of Unexpected Dust Destruction using High-Redshift Post-Starburst Galaxies (PI: J. Spilker)

PUBLICATIONS

First-Author

3. **D’Onofrio, V.**, et al. (2026). “Radio AGN Candidates in Young, Gas-rich SQuIGGLE Post-starburst Galaxies at $z \sim 0.7$.” Under collaboration review.
2. **D’Onofrio, V.**, et al. (2026). “Molecular Gas Excitation in $z \sim 0.7$ Gas-Rich Post-starburst Galaxies from SQuIGGLE.” ApJ, 1003, 52.
1. **D’Onofrio, V.**, et al. (2025). “Quenching Through Tidal Gas Removal: Molecular Gas and Star Formation in Tidal Tails of $z \sim 0.7$ Post-Starburst Galaxies.” ApJ, 990, 166.

Co-Author

8. Knowlton, M., et al. (**incl. D’Onofrio, V.**) (2026). “A Possible Protocluster of Galaxies Serendipitously Discovered in the Field of an Intermediate-redshift Post-starburst Galaxy.” RNAAS, 10, 102.
7. Luo, Y., et al. (**incl. D’Onofrio, V.**) (2026). “A Multiwavelength Evaluation of AGN in the Post-Starburst Phase.” ApJ, 1000, 24.
6. Setton, D., et al. (**incl. D’Onofrio, V.**) (2025). “SQuIGGLE: Buried star formation cannot explain the rapidly fading CO(2–1) luminosity in massive, $z \sim 0.7$ post-starburst galaxies.” AJ, 170, 351.
5. Spilker, J., et al. (**incl. D’Onofrio, V.**) (2025). “Unusually High Gas-to-Dust Ratios Observed in High-Redshift Quiescent Galaxies.” ApJL, 993, L40.
4. Kumar, A., et al. (**incl. D’Onofrio, V.**) (2025). “Meet the Neighbors: Gas Rich “Buddy Galaxies” are Common Around Recently Quenched Massive Galaxies in the SQuIGGLE Survey.” RNAAS, 9, 243.

3. Suess, K., et al. (**incl. D'Onofrio, V.**) (2025). "Cold gas in a post-starburst pair at $z \sim 1.4$: major mergers as a pathway to quenching in the HeavyMetal survey." ApJ, 993, 158.
2. Zhu, P., et al. (**incl. D'Onofrio, V.**) (2025). "SQuIGGLE: Observational Evidence of Low Ongoing Star Formation Rates in Gas-rich Post-starburst Galaxies." ApJ, 981, 60.
1. Verrico, M., et al. (**incl. D'Onofrio, V.**) (2023). "Merger Signatures are Common, but not Universal, in Massive, Recently Quenched Galaxies at $z \sim 0.7$." ApJ, 949, 5.

TALKS

Invited

4. **D'Onofrio, V.**, et al. "Investigating Galaxy Quenching through Radio Observations of Post-Starburst Galaxies," Tuesday UVa / NRAO Astronomy (TUNA), Charlottesville, VA, November 3, 2026.
3. **D'Onofrio, V.**, et al. "Investigating Galaxy Quenching through Radio Observations of Post-Starburst Galaxies," Princeton University GalRead, Princeton, NJ, October 29, 2026.
2. **D'Onofrio, V.**, et al. "Investigating Galaxy Quenching through Radio Observations of Post-Starburst Galaxies," STScI/JHU Galaxy/AGN Journal Club, Baltimore, MD, October 28, 2026.
1. **D'Onofrio, V.**, et al. "Tidal Removal of Cold Molecular Gas in $z \sim 0.7$ Post-Starburst Galaxies: A Qualitatively New Quenching Mechanism," Texas A&M Nuclear-Astro Seminar, College Station, TX, May 24, 2024.

Contributed

10. **D'Onofrio, V.**, et al. "Insights from the Radio into Star Formation Suppression in SQuIGGLE Post-Starburst Galaxies," Texas A&M Astrosymposium, College Station, TX, August 21, 2026.
9. **D'Onofrio, V.**, et al. "Investigating Star Formation Suppression via Higher-Redshift Post-Starburst Galaxies," The End of Star Formation, Urbana-Champaign, IL, March 3, 2026.
8. **D'Onofrio, V.**, et al. "Investigating Star Formation Suppression via the ISM of Higher-Redshift Post-Starburst Galaxies," Decoding Galactic Evolution through the Interplay of the Multi-Phase Interstellar Medium, Nagoya, August 25, 2025.
7. **D'Onofrio, V.**, et al. "Tracing galaxy quenching via post-starbursts with recent observations from the radio," Texas A&M Astrosymposium, College Station, TX, August 18, 2025.
6. **D'Onofrio, V.**, et al. "Probing Star Formation Suppression with Higher- z Post-Starbursts," Texas A&M Astrosymposium, College Station, TX, August 15, 2024.

5. **D’Onofrio, V.**, et al. “Tidal Removal of Cold Molecular Gas in $z \sim 0.7$ Post-Starburst Galaxies: A Qualitatively New Quenching Mechanism,” McMaster University Star Formation Workshop, Hamilton, ON, August 14, 2024.
4. **D’Onofrio, V.**, et al. “Tidal Removal of Cold Molecular Gas in $z \sim 0.7$ Post-Starburst Galaxies: A Qualitatively New Quenching Mechanism,” STScl Spring Symposium 2024, Baltimore, MD, April 19, 2024.
3. **D’Onofrio, V.**, et al. “Quenching in Intermediate-Redshift Massive Post-Starburst Galaxies,” Texas A&M Astrosymposium, College Station, TX, August 17, 2023.
2. **D’Onofrio, V.**, et al. “Molecular Gas Toward Sagittarius B2: The Galactic Center’s Complex Giant Molecular Cloud,” Texas A&M Astrosymposium, College Station, TX, August 18, 2022.
1. **D’Onofrio, V.**, et al. “Characterizing Molecular Gas Towards Sagittarius B2: The Galactic Center’s Complex Giant Molecular Cloud,” 2021 NRAO/GBO Summer Student Symposium, virtual participation, August 13, 2021.

Public

2. **D’Onofrio, V.** “A Day in the Life of an Astronomy PhD Student,” TAMU High School Outreach Event, College Station, TX, November 12, 2025.
1. **D’Onofrio, V.** “Why You Shouldn’t Go to Mars,” Astronomy on Tap, Bryan, TX, August 23, 2023.

Other

4. **D’Onofrio, V.**, et al. “Evolution Through the Post-starburst Phase,” Texas A&M Exgal Group Meeting, College Station, TX, October 20, 2025.
3. **D’Onofrio, V.**, et al. “Tidal Removal of Cold Molecular Gas in $z \sim 0.7$ Post-Starburst Galaxies: A Qualitatively New Quenching Mechanism,” Master’s/Preliminary Exam Defense, College Station, TX, October 28, 2024.
2. **D’Onofrio, V.**, et al. “Star Formation Quenching in Intermediate-Redshift Massive Post-Starburst Galaxies,” Texas A&M Exgal Group Meeting, College Station, TX, April 11, 2024.
1. **D’Onofrio, V.**, et al. “Molecular Gas Toward Sagittarius B2: The Galactic Center’s Complex Giant Molecular Cloud,” MSU Society of Physics Students Coffee Breaks, East Lansing, MI, November 11, 2021.

LEADERSHIP & OUTREACH

STEM Pen Pal
Letters to a Pre-Scientist

August 2025 – Present
 Texas A&M University

Coordinator <i>Mentoring and Advising Graduates in an Inclusive Community (MAGIC)</i>	February 2025 – Present Texas A&M University
Coordinator <i>Graduates Learning Astro and Soft Skills (GLASS)</i>	October 2023 – February 2025 Texas A&M University
In the News Presenter <i>Astronomy on Tap</i>	October 2023 – April 2026 Bryan, TX
Mentor <i>Mentoring and Advising Graduates in an Inclusive Community (MAGIC)</i>	August 2023 – August 2026 Texas A&M University
Trivia Creator <i>Astronomy on Tap</i>	May 2023 – July 2024 Bryan, TX
Astronomer/Physicist <i>Adopt-a-Physicist</i>	October 2023 – Present Texas A&M University

STUDENT MENTORING

Aspen Presswood — TAMU undergraduate Radio variability in SQuGGLE galaxies with FIRST/VLASS.	Spring 2026
Mary Knowlton — TAMU undergraduate; AggieSTAAR mentee Dust continuum observations of gas-rich SQuGGLE galaxies; <u>publication</u> in RNAAS. Now a graduate student at Indiana University Bloomington.	Summer 2025

TEACHING EXPERIENCE

Teaching Assistant, Lab Instructor <i>ASTR 111: Overview of Modern Astronomy</i>	January 2024 – May 2024 Texas A&M University
Teaching Assistant <i>ASTR 320: Astrophysical Research Methods</i>	August 2023 – December 2023 Texas A&M University
Teaching Assistant, Lab Instructor <i>ASTR 111: Overview of Modern Astronomy</i>	January 2023 – May 2023 Texas A&M University
Teaching Assistant <i>ASTR 401: Stars and Extrasolar Planets</i>	August 2022 – December 2022 Texas A&M University

OBSERVATION EXPERIENCE

MSU Observatory 0.6-meter Telescope, **30 hrs**

REFERENCES

Justin Spilker — Assistant Professor, Texas A&M University
Ph.D. thesis advisor | jspilker@tamu.edu

Krista Lynne Smith — Associate Professor, Texas A&M University
Ph.D. thesis committee member | kristalynnesmith@tamu.edu

Wren Suess — Assistant Professor, University of Colorado Boulder
Research collaborator | suess@colorado.edu